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COMMERCIAL ITEM DESCRIPTION

DUAL TRANSMISSION MEDIA DATA COMMUNICATION SYSTEM

(DTMDCS)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers a data communication system for Intrusion Detection Systems (IDS) used to protect sensitive conventional arms, ammunition and explosives (AA&E). The DTMDCS uses dual transmission media (telephone link with backup cellular radio link) to assure successful notification of alarm and status change conditions.

1.1 Intended use. The DTMDCS is intended to provide IDS alarm and system status data communications from one or more control units (also referred to as remote area data collectors, alarm panels, alarm controls, premise control units, etc.) which are located in the protected area, to a remotely located monitor console (Central Station, remote status monitor, etc.).

NOTE: The Control-Communicator, Digital described herein could be used as a stand-alone device without requiring another control unit; however, the intended application is for use with existing control units of the Joint-Services Interior Intrusion Detection System (J-SIIDS). The J-SIIDS Control Unit has three outputs: alarm or not alarm, secure or access (armed or disarmed), and AC power present or absent.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent to: USA Communications-Electronics Command, Logistics & Readiness Center, ATTN: AMSEL-LC-LEO-E-EB, 10115 Gridley Rd, Suite 101, Ft Belvoir, VA 22060-5856.

AMSC N/A FSC 6350

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2. SALIENT CHARACTERISTICS

2.1 Environmental requirements. The components shall operate as specified herein at temperatures from 32 to 120 °F and at relative humidities up to 85 (±5) percent.

2.2 Major components.

a. Control-Communicator, Digital (hereinafter called “DCC”)
b. Keypad, Control-Communicator, Programming (hereinafter called “keypad”)
c. Expansion Module, Zone, Control-Communicator (hereinafter called “expansion module”)
d. Transmission Device, Alarm, Cellular (hereinafter called “CATD”)

Note: The DCC and CATD may be separate components which work together in an interactive system or a single integrated component.

2.3 Control-Communicator, Digital:

2.3.1 Primary function. The primary function of the DCC shall be to monitor the outputs of external devices for the changes specified herein and provide notification of these changes to one or more remotely located monitoring consoles by use of the public switched telephone network. Each device output (also referred to as a zone) will present one of the following conditions: (a) normally open relay contacts which close on alarm or change of status; (b) normally closed relay contacts which open on alarm or change of status.

2.3.2 Number of zones. The DCC shall support a minimum of eight fully programmable hardwired zones and be expandable to support at least fifty-six additional zones. The additional zones may be connected to the DCC by additional hardwired zone inputs within the DCC or by means of a multiplexed polling loop using one or more expansion modules (see 2.3.23) to which the device outputs are hardwired. Any hardware component required to implement the polling loop, other than the expansion modules, shall be integral to the DCC or delivered with the DCC designed for installation therein.

2.3.3 Zone response time. The zone response time (the time a change at one of the device outputs must persist before the DCC will communicate the change) shall be not less than 100 and not greater than 500 milliseconds.

2.3.4 Line supervision. Line supervision between the DCC and device outputs shall be accomplished using current supervision through end-of-line (EOL) resistors located at the device outputs or by any other line supervisory technique which provides equal or greater line security.

2.3.5 Partitions. The DCC shall be configurable into a minimum of 6 partitions. Partitions shall function like separate premises, with separate zones and users. Each zone shall be assignable to any of the partitions. Users shall be able to perform system functions, such as arming and

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disarming, in any selected partition, if authorized, without affecting other partitions. The system functions the user may initiate shall depend on the user's assigned authority level in that partition (see 2.3.14). One or more keypads (see 2.3.22) shall be utilized to initiate these system functions. Any partition shall be controllable from any keypad. When any keypad is being operated, data entered shall not be displayed on any other keypad. User security codes shall never be displayed.

2.3.6 Reports. The DCC shall report to the telephone number(s) programmed (see 2.3.12) when a device output changes from its normal state for the minimum duration specified herein (see 2.3.3) and when system events or user initiated system functions occur that are programmed for transmittal of reports. The DCC shall be programmable to send reports when the device outputs return to the normal state (restorals). Reports shall be transmitted by digital code. The transmission data format(s) shall be compatible with various commonly used existing central station monitors and shall not be limited to one specific format. The DCC shall be programmable to transmit reports in a different communications format to each telephone number. The DCC shall be capable of conveying, as a minimum, the following information to the extent applicable: subscriber account number, the zone input that has changed and whether it is an alarm (that is, change from normal state) or restoral, type of system event or system function and user initiating the system function.

2.3.7 Operating power. The DCC shall operate from nominal 60 Hertz, 115 volts alternating current (Vac), wired directly to the DCC or from the output of an external transformer which plugs into a nominal 115 Vac outlet. The DCC shall perform its normal signaling functions when energized by voltages from 85 to 110 percent of the nominal voltage.

2.3.8 Audible signaling device power. The DCC shall provide either or both of the following outputs for operating an audible signaling device: (a) a nominal 12 volt dc (Vdc), 1 ampere minimum, output for operating a bell, siren, or other audible signaling device; or (b) a signal at an amplitude of 1.2 amperes minimum into an 8 ohm external speaker. If option (b) is provided, both a steady tone and either a warbling or yelping tone shall be provided and shall be user selectable. The DCC may draw supplementary power from the standby battery when the audible signaling device is sounding. The output to the audible signaling device shall be initiated by receipt of an alarm condition at a zone input programmed to initiate this output. The duration of this audible output shall be programmable. As a minimum, durations of 5 ±1, 10 ±1, 15 ±1, and 30 minutes or greater shall be selectable.

2.3.9 Auxiliary and standby power. The DCC shall provide auxiliary power of 12 to 14 Vdc at a current of at least 500 milliamperes (mA) for operation of keypads, external sensors, expansion modules, etc. The DCC shall house and maintain on float or trickle charge an internal rechargeable, lead acid battery, or batteries, capable of operating the DCC for a minimum of eight hours, while delivering the auxiliary current of 500 mA with no alarms. After restoration of 115 Vac power and while in normal operation in the non-alarm state, the DCC shall recharge the battery sufficiently within 24 hours to provide at least 4 hours of battery operation and 8 hours after 72 hours of charging.

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2.3.10 Operation on ac power failure. Upon loss of ac power, the DCC shall immediately operate from battery power. Loss or subsequent restoral of ac power shall not cause a false alarm. The DCC shall, if programmed, initiate a report of the loss of ac power to the phone number(s) dialed (see 2.3.12) not less than 30 seconds nor more than 40 minutes after ac power loss occurs. No report shall be sent if power is restored prior to transmittal of the report. The keypad shall indicate loss of ac power not more than 120 seconds after ac power loss.

2.3.11 Low battery indications. When the DCC is operating from standby power, the battery voltage shall be monitored, and a report indicating a low battery shall be transmitted before the DCC becomes unable to perform its functions. When the DCC is operating from ac power, the DCC shall check the battery condition under load at least daily and shall transmit a low battery signal if the battery would be unable to properly perform its standby function. A low battery indication shall also be made at the keypad.

2.3.12 Dialing modes. The DCC shall be programmable to transmit reports of changes at its zone inputs, system status changes and user operations to at least two telephone numbers. As a minimum, the DCC shall provide the following dialing mode options:

a. transmit reports to the primary phone number; if the transmission attempt to this number is unsuccessful after a preset or programmed number of dialing attempts, transmit to a second number (back-up reporting);
b. transmit reports to the primary number, then to the secondary number (dual reporting). In this mode and in mode a. the DCC may alternate dialing between the two numbers after one or two unsuccessful attempts to the alternate number;
c. transmit zone change reports only to the primary phone number and all reports (including zone change reports) to the secondary phone number.

If attempts to report are unsuccessful, a total of at least 8 attempts shall be made per phone number before dialing attempts are discontinued. An indication of the communications failure shall then be made at the keypad.

2.3.13 Programming. The DCC shall be fully programmable in the field with the same keypad used for system control (see 2.3.22). Programming with the keypad shall require entry of a programmable four, or greater, digit code (installer code). It shall also be possible to download and upload program data and to perform primary user control functions from a remote location. The DCC shall incorporate an optional download/upload callback technique. When programmed for this option, after receiving an initial call from a remote location requesting downloading/uploading, the DCC shall be required to call a preprogrammed phone number from which the downloading/uploading can proceed.

2.3.14 User operations. To perform arming, disarming and other system functions, it shall be necessary to enter a four or more digit code (user code). At least 32 user codes shall be

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assignable. Operations (that is, system functions) a given user may perform shall be determined by the authority level assigned to that user. A master user shall be capable of performing all possible system functions and assigning other user codes and authority levels, except the master level. As a minimum, other authority levels shall include the following: one that allows the user only to arm and disarm the DCC, but not to perform other system functions such as bypassing zones, viewing the log, etc.; one that allows the user only to arm the DCC, or, one that allows the user to arm the DCC and also to disarm it if that user was the person who armed it. The DCC shall allow a user to transmit a duress alarm report by appropriate keypad entry without annunciation of an alarm at the keypad and external sounder.

2.3.15 Zone types. The DCC shall provide, as a minimum, two types of zones, a 24-hour zone and a delay zone. For zones programmed as 24-hour zones, changes at the zone inputs shall be immediately reported to the monitor station regardless of the arming status. For zones programmed as delay (entry/exit) zones, an alarm occurring when the system is disarmed shall be ignored. The DCC shall be armed by keypad entry of an appropriate user code and keystroke or keystrokes. Arming the DCC shall initiate the exit delay period. Alarms received during this period shall be ignored. After expiration of the exit delay period, receipt of an alarm shall initiate the entrance delay period. If the DCC is disarmed by a user before expiration of the entrance delay period, alarms received during the entrance delay period shall be ignored. If the DCC is not disarmed before expiration of the entrance delay period, an alarm report shall be transmitted to the monitor station(s) and local annunciation shall occur if programmed. The entrance and exit delay periods shall be independently programmable. At least two sets of delay times shall be selectable. Selectable times shall range from 15 seconds or less to at least 150 seconds in increments no greater than 15 seconds.

2.3.16 Real time clock. The DCC shall contain a real time clock, settable at the keypad or remotely by downloading.

2.3.17 System event log. The DCC shall store, as a minimum, the most recent 100 alarms, system events, and user initiated system functions. These events shall be viewable at the keypad and from a remote computer. Displays shall include, as applicable, the type of event (alarms, restorals, openings, closings, etc.), zone number, partition, user number, and date and time of occurrence.

2.3.18 Non-volatile storage. All program information stored in the DCC shall be stored in non-volatile memory.

2.3.19 Periodic self test. As a minimum, the DCC shall be capable of automatically initiating a self test signal to verify the communications link at selectable intervals of once per day and once per week. The time of day when the initial test signal is sent shall be selectable to within an hour.

2.3.20 Test of CATD. The DCC shall be able to process a separate test signal from the CAT to verify the integrity of the cellular link.

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2.3.21 UL compliance. The DCC shall be designed and manufactured to comply with Underwriter's Laboratories, Inc. (UL) Standard 1635, when installed in accordance with UL requirements, except as specified herein. The DCC need not be certified by UL. Unless otherwise specified in the solicitation or purchase order, the DCC need not meet the requirement for sending an alarm or trouble signal when removed from its installed location and the requirement for a tamper switch on the door or cover.

2.3.22 Keypad. The keypad shall be the principle means of programming and control of the DCC. It shall contain, as a minimum, keys numbered 0 through 9 and any other keys necessary to access programmable system variables and to enter data therein during programming and to initiate system functions. The keypad shall incorporate a visual display to facilitate programming and indicate alarms and system events. The keypad shall incorporate an audible sounder that shall sound on alarms and other events if programmed. The keypad shall incorporate all features required to enable programming and operation of the DCC as specified in this CID.

2.3.23 Expansion modules. Expansion modules shall provide hardwire input terminals for two or more device outputs conforming to 2.3.1. The expansion modules shall recognize changes in the device output conditions and shall communicate the changes when polled by the DCC. The expansion modules shall receive operating power from the DCC or from a separate 12 Vdc power source. It is desired that removal of the cover shall cause a normally closed condition across a pair of terminals to go to the open state. All other characteristics required of the modules to operate with the DCC in the manner required to meet the requirement of this CID to monitor up to 64 zones, shall be incorporated therein. Unless otherwise specified in the contract or order (see 6.1), an expansion module shall provide inputs for two device outputs.

2.4 Cellular Alarm Transmission Device

2.4.1 Function. The CATD shall provide a secondary data link over cellular radio when it determines the public switched telephone network data link is inoperative.

2.4.2 Telephone line fault monitoring. The CATD shall monitor the telephone line voltage and, upon detection of changes that indicate a line fault on the telephone link, the CATD shall automatically switch to the cellular link and transmit any messages generated by the DCC over the cellular link to the phone number(s) dialed by the DCC. This switch to cellular transmission shall occur not more than 15 seconds after a line fault is detected. It is desired that, if a transmission is initiated by the DCC before the CATD has switched to cellular, the switch to cellular operation shall occur with no delay. The CATD shall switch back to communication via the telephone link not more than 2 minutes after restoral of the telephone link. The CATD shall also provide a relay output, indicating the line fault condition, not less than 8 seconds nor more than 60 seconds after detection of the line fault.

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2.4.3 Cellular link monitoring. The CATD shall monitor its communication status and, on determination of loss of contact with its cell site or other loss of communications capability, shall provide a relay output, distinct from the line fault output (see 2.4.2), indicating loss of cellular link.

2.4.4 Periodic self test . The CATD shall provide a periodic self test at selectable weekly or monthly intervals. To perform this test, the CATD shall provide a relay output to trip an input to the DCC and transmit a test report over the cellular link to the phone number(s) dialed (see 2.3.12) by the DCC. This relay shall be dedicated only for the periodic self test and any manually initiated self test and shall be distinct from the line fault (see 2.4.2) and cellular link loss (see 2.4.3) relays. Not more than 2 minutes after completion of a self-test, the CATD shall switch back to communication via the telephone link.

2.4.5 Operating power. The CATD shall operate from nominal 115 volts, 60 Hz alternating current wired directly to the CATD or from the output of an external transformer which plugs into a nominal 115 volts ac outlet. The CATD shall perform its normal functions when energized by voltages from 85 to 110 percent of the nominal voltage.

2.4.6 Standby power. The CATD shall maintain on float or trickle charge an internal rechargeable, lead acid battery, or batteries, which shall, upon failure of the ac power, automatically provide operating power for the CATD for at least 8 hours, assuming at least fifteen minutes of cellular transmissions. The CATD shall provide a relay output that changes state to indicate loss of ac power. The change in this output shall not occur if ac power is lost for less than one minute but shall occur after not more than 4 hours without ac power, except as follows: the time may be as much as 8 hours if the calculated battery capacity remaining when the change of state occurs is sufficient to allow battery operation for at least 4 more hours, followed by at least 5 transmission attempts at maximum (3 watts) output power. The CATD shall check battery condition at least once daily and shall provide a relay output to indicate low battery if the capacity has dropped to a level that would render the battery unable to perform its function. The same relay may be used to indicate loss of ac power, low or missing battery, and loss of cellular link and other CATD problems, but must be different from the relays indicating telephone line fault (see 2.4.2) and self test (see 2.4.3).

2.4.7 Output power. The CATD shall be able to deliver up to 3 watts output power to a standard 5/8 wave, 14.5 inch mast length, cellular antenna with minimum of 10 feet of coaxial cable.

2.4.8 Dialing attempt monitoring. The CATD shall employ a method of determining unsuccessful DCC dialing attempts and shall switch to cellular transmission if the DCC does not successfully contact the monitor station after a predetermined pattern of dialing attempts which indicates failure of the DCC to successfully transmit its report. Provision shall be made for the user to disable this option.

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2.4.9 UL compliance. The CATD shall be designed and manufactured to comply with Underwriter's Laboratories, Inc. (UL) Standard 1635, when installed in accordance with UL requirements, except as specified herein. The CATD need not be certified by UL. Unless otherwise specified in the solicitation or purchase order, the CATD need not meet the requirement for sending an alarm or trouble signal when removed from its installed location and the requirement for a tamper switch on the door or cover.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 The components shall be certified to conform to all applicable FCC regulations for standard telephone and cellular phone service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 The item shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

a. Salient characteristics not as specified.
b. Preservation, packing, and marking not as specified in the contract or order.

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 The procuring agency should include the following information and applicable requirements in procurement documents:

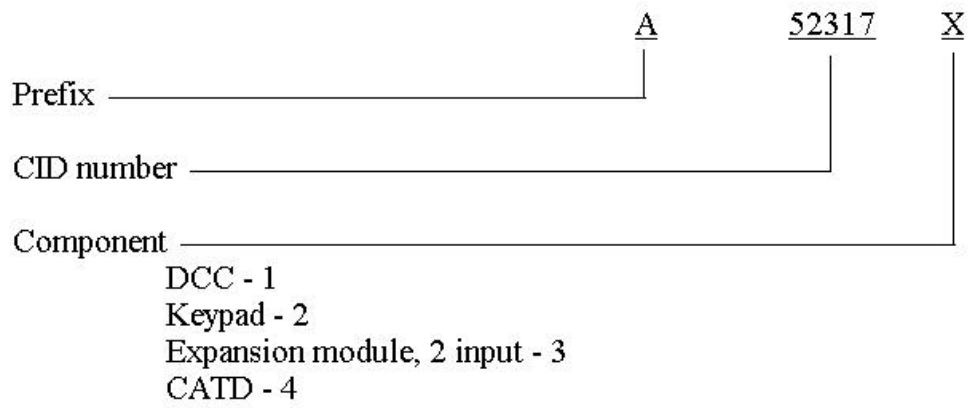
a. Title, number and date of this commercial item description.
b. Nomenclature and quantity of component or components required.
c. Number of inputs required on expansion modules if other than two.

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d. When this commercial item description is used for procurement, the commercial item description clause must appear in the solicitation.
e. Solicitation should specify that all auxiliary parts or components required for normal operation, in accordance with section 2, of the items being procured, such as transformers, end-of-line resistors, antennas, etc. (except batteries) are to be supplied and packed with each item they are required to support.
f. Contract should include statement that operator's manual, programming manual and installation manuals, or instructions, as appropriate, shall be supplied and packed with each unit.

6.2 CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

6.2.1 Part or identifying number (PIN). The PINs to be used for components acquired to this commercial item description are created as follows:



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6.3 The following components are believed to meet the requirements of this CID. However, the list is not exhaustive and competition should not be limited to these manufacturers. Contract documents should state that any product meeting the requirements of this CID will be equally considered.

Digital Control-Communicator

Alarm Device Manufacturing Company (ADEMCO) 163 Eileen Way Syosset, NY 11791	models Vista 50-P or 50-PUL
Europlex Technologies (U.S.A.) Inc. 1700 Enterprise Way, Suite 106 Marietta, GA 30067	model ADPLEX 2064, PAK04
Fire Burglary Instruments, Inc. 165 Eileen Way Syosset, NY 11791	models XL-4 or XL-4B, plus model 7120 multiplex expander module

Keypad

ADEMCO	model 6139
Europlex Technologies (U.S.A.), Inc.	model 2121
Fire Burglary Instruments, Inc.	model XK7LC

Expansion module

ADEMCO	Model 4190WH
Europlex Technologies (U.S.A.), Inc.	Model ADLEM
Fire Burglary Instruments, Inc.	Model XL-4702

Cellular Alarm Transmission Device

Cellular Alarm Products, LTD. 2575 Southwell, Suite 104 Dallas, TX 75229	Model HS1 Deluxe Wizzard with Monthly Timer
Europlex Technologies (U.S.A.), Inc.	Model: ACT-1 Kit (see note below)
Telular Corporation 4130 Shirley Drive SW Atlanta, GA 30336	Model T-1630M Model T-1630MA Model T-1110M Model T-1210M Model T-3100UL.

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Note: Europlex ACT-1 kit works only with Europlex DCC.

6.4 The preparing activity for this CID is: USA Communications-Electronics Command, Logistics & Readiness Center, ATTN AMSEL-LC-LEO-E-EB, 10115 Gridley Rd, Suite 101, Ft Belvoir, VA 22060-5856.

MILITARY INTERESTS:

Custodians
Army - CR4
Navy - SH
Air Force - 99

Review activities
DLA - CC, GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing activity
Army - CR4

Project 6350-0225

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